Bright Coffee Shop Sales Analysis

Executive Presentation

Executive Context & Objectives

Executive Overview & Business Context

This presentation provides an executive-level view of sales performance at Bright Coffee Shop, using historical transaction data to support strategic decisions. With a new CEO focused on revenue growth, stronger product performance, and a more data-driven culture, this work translates raw sales records into actionable insight.

The dataset captures daily transactions including product details, quantities, pricing, and time of sale. The analysis is organized around business questions: (1) which categories and items drive revenue, (2) when peak and slow periods occur, (3) how sales trend over time, and (4) what recommendations leadership can implement immediately.

Rather than emphasizing isolated metrics, the focus is on patterns, comparisons, and trade-offs that inform operational, marketing, and product decisions. The narrative moves from objectives and method to key findings, then concludes with clear next steps to improve financial performance and operational efficiency.

Business Objectives & Key Questions

Focused analysis tied to CEO growth goals

Why this analysis exists

This project is designed to answer a defined set of business questions that directly support revenue growth and performance improvement at Bright Coffee Shop. Every KPI, chart, and insight in the deck maps back to these objectives.

Objectives guiding the work

1) What products generate the most revenue?

Identify top categories (e.g., beverages vs. food) and top individual items to inform promotion, expansion, and rationalization decisions.

2) When does the store perform best?

Group transactions into standardized time intervals to surface peak periods and underperforming slots for staffing, hours, and campaign timing.

3) What are the sales trends by product and time?

Track revenue and volume patterns across intervals and days to understand demand cycles and spot anomalies worth investigating.

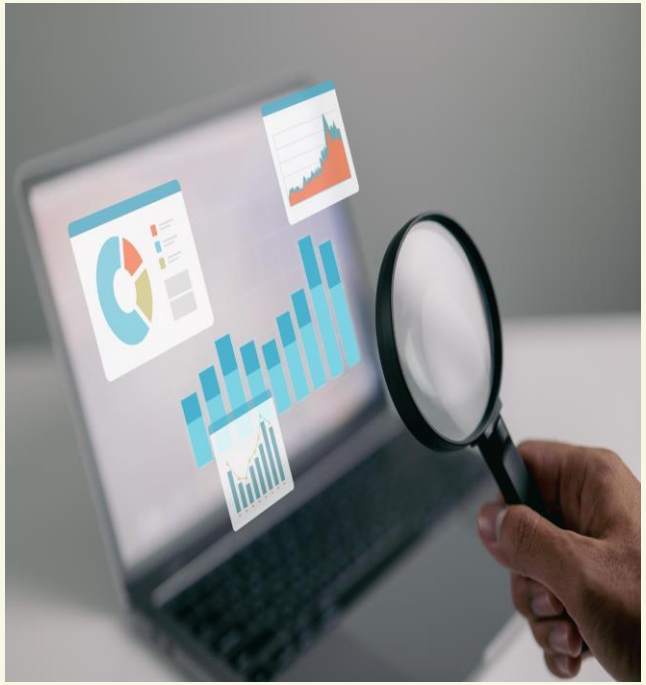
4) What actions will improve performance?

Turn findings into practical recommendations (e.g., time-based promotions, inventory optimization, and future automation opportunities).

Data, Tools, and Methodology

Dataset Overview & Preparation

Cleaning, standardizing, and enriching POS records



Dataset structure

Bright Coffee Shop Sales captures item-level POS transactions (date, time, category, product detail, unit price, quantity). This granularity supports both revenue KPIs and behavioral patterns across the day.

Cleaning & standardization

The raw Excel file required fixes for numeric formatting (e.g., comma vs. decimal unit prices) and consistent time parsing. Standardized types prevent distorted revenue totals and incorrect time-based trends.

Derived fields for analysis

New metrics included total transaction revenue (unit price × quantity) and time buckets (30-minute or 3-hour intervals) to support operational decisions like staffing and promo timing.

Result: a clean, structured dataset ready for reliable aggregation, visualization, and executive reporting.

Analytical Tools and Workflow

Databricks: processing and transformation

SQL was used to clean data, create calculated fields, and aggregate results across products and time intervals—keeping transformations transparent, repeatable, and scalable.

Excel: analysis and visualization

The analytical table was exported for pivot tables and pivot charts to compute KPIs (revenue by category, units by detail, and revenue by time bucket) with consistent numbers and visuals.

Miro + PowerPoint: plan and communicate

Miro clarified data flow and architecture; PowerPoint assembled results into a narrative executive deck—prioritizing clarity, reproducibility, and business alignment.

Sales Performance Analysis

Revenue by Product Category

Top Revenue Category: Coffee

What this means?
Coffee generates the highest total revenue compared to all other product categories. This indicates strong and consistent customer demand, likely driven by repeat purchases, brand familiarity, and daily consumption habits.

Revenue Concentration: High

What this means?

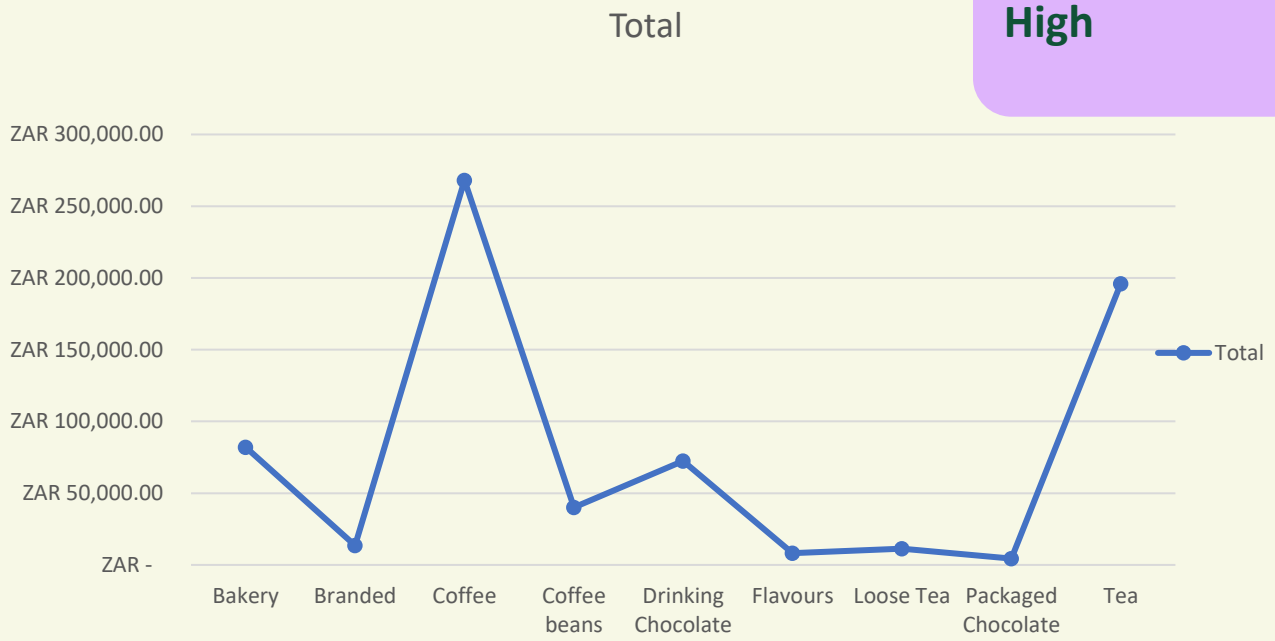
A large share of total revenue comes from a small subset of product categories, rather than being evenly distributed across the menu.

Why this matters?
The business is not reliant on wide diversification for revenue.
Risk and opportunity are both concentrated:
Risk: over-reliance on a few categories
Opportunity: focused optimisation has high impact

<u>Business implication</u>
Leadership should focus strategic attention where revenue is concentrated, while applying a clear framework to evaluate whether low-contributing categories justify their complexity and cost.

Top Revenue Category
Coffee

Revenue Concentration
High



Product-Level Sales & Best Sellers

Revenue shows a consistent month-on-month increase across all store locations from January to June.

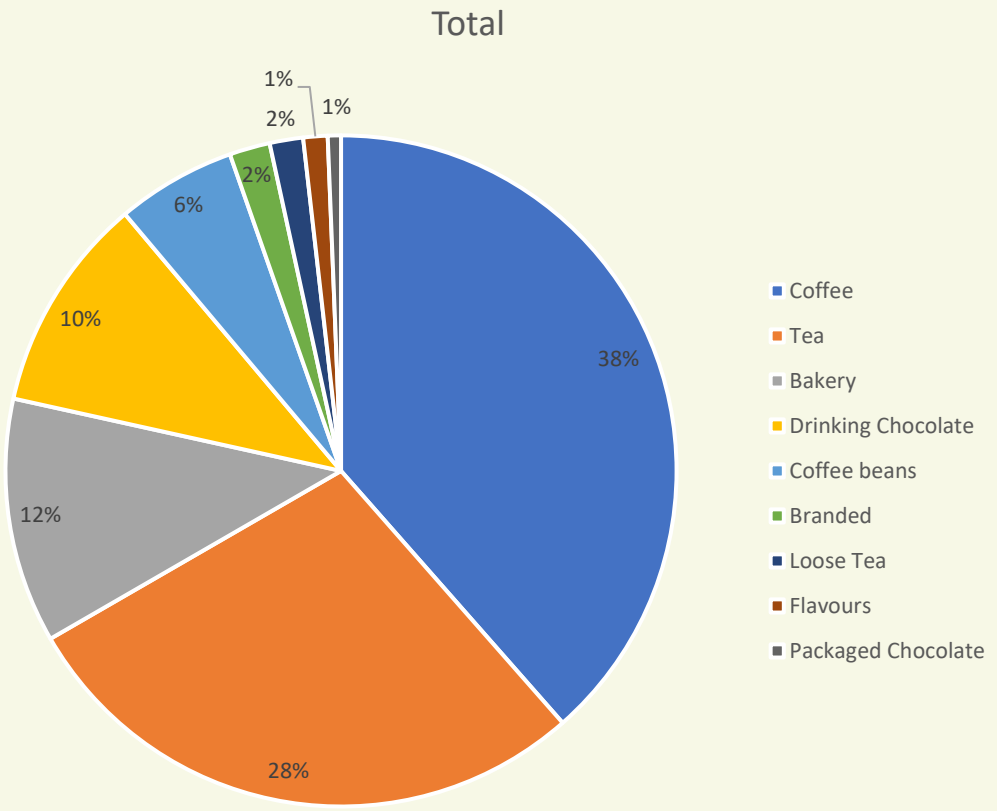
Hell's Kitchen is the strongest revenue contributor, particularly from March onwards. Lower Manhattan closely follows Hell's Kitchen, indicating strong and sustained growth.

Astoria records lower revenue but mirrors the same positive growth pattern.

What This Means for the Business?
Growth is broad-based, not dependent on a single store.

Revenue trends are stable and predictable, supporting confident planning and forecasting.

High-performing locations can serve as benchmarks for operational and marketing best practices.



Best Seller
Coffee & Tea

Low Performers
Packaged Items

Time-Based Sales Performance

Peak Time: Morning

What this means:

The morning time bucket consistently generates the highest revenue across all locations, outperforming afternoon and evening periods.

Why this matters? Morning demand is predictable and repeatable.

Customers in this period are often time-sensitive and convenience-focused

Service speed and availability directly affect revenue capture

Business implication: Staffing, inventory preparation, and service workflows should be optimized for morning peaks to avoid lost sales and customer dissatisfaction.

Slow Time: Evening

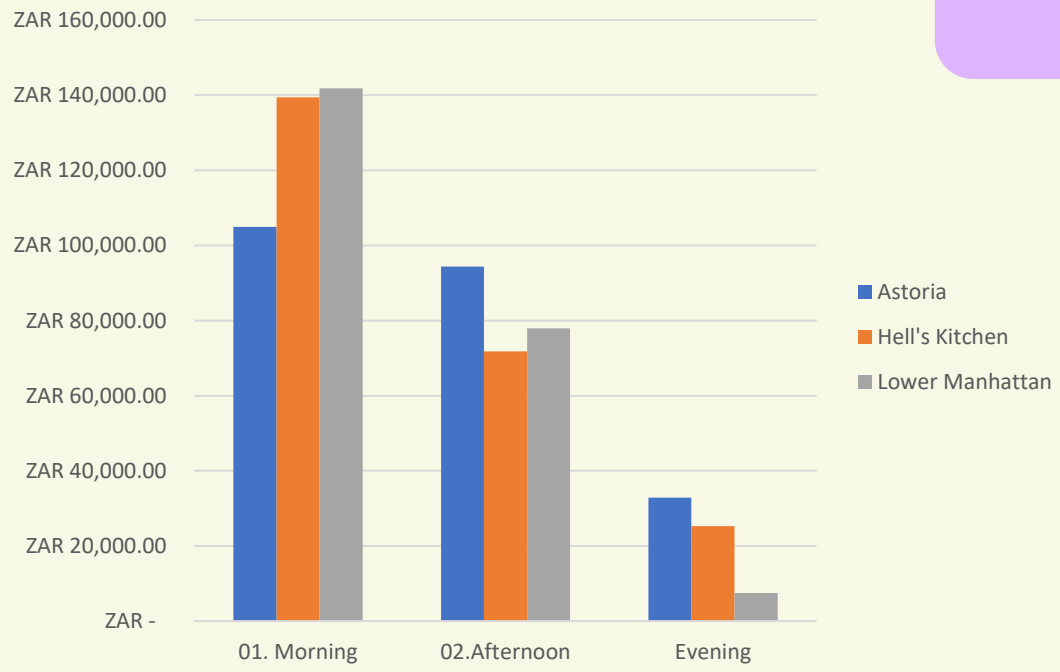
What this means? Evening sales consistently underperform, contributing the least revenue of all time buckets across locations.

Why this matters? Uniform staffing or stock levels throughout the day may be inefficient.

Evening demand may not justify peak-level operating costs

This time window represents either ,A cost-control opportunity Or a targeted promotion opportunity.

Business implication : Leadership can either reduce evening operational intensity to control costs, or test time-specific promotions (e.g., bundles or discounts) to stimulate demand without increasing fixed expenses.



Peak Time

Morning

Slow Time

Evening

Insights and Recommendations

Key Insights from the Analysis

Concentration: Sales are not evenly distributed across products or time periods. A small subset of products and categories drives a large share of total revenue, underscoring the need for focus and prioritization.

Rhythms: Customer demand follows consistent daily patterns with repeatable peak and slow periods. These patterns provide a reliable baseline for planning.

Volume vs. value: High sales volume does not always align with high revenue. Some items sell many units at low price, while others deliver strong revenue with fewer units—critical for pricing and promotion decisions.

Operational alignment: Uniform staffing or stock levels can be inefficient during slow periods and insufficient during peaks. Using evidence-based adjustments can improve customer experience and cost efficiency, reducing reliance on intuition and lowering decision risk.

CEO Recommendations

Practical steps to drive revenue and performance

1) Protect top performers

Ensure consistent availability and quality for best-selling products and categories. Where demand is strong and sensitivity is low, test modest price optimization to lift revenue.

2) Use time-based marketing

Introduce slow-period offers (limited-time discounts, bundles, or loyalty rewards) to shift visits outside peak hours. This improves utilization without materially increasing fixed costs.

3) Align staffing and inventory to demand

Staff and prep more during peaks to reduce wait times and prevent lost sales. Use leaner coverage during slow periods to control costs while maintaining service quality.

4) Review underperformers with a framework

Evaluate low performers by revenue contribution, strategic value, and improvement potential. Reposition or promote where viable; remove items that cannot justify shelf space.

Future Outlook

Next Steps & Automation Opportunities

Automate reporting: Build daily/weekly dashboards that refresh from source data to reduce manual effort and keep metrics current.

Scale to multiple locations: Standardize KPIs and reporting to compare stores, share best practices, and address underperformance.

Add customer-level data: Use loyalty participation to analyze repeat behavior and lifetime value, enabling targeted promotions.

Forecast demand: As history grows, introduce predictive analytics to anticipate sales patterns and optimize inventory planning.

These steps extend the project into an ongoing foundation for analytical maturity and operational excellence.